Year 10 Foundation

REVISION PACK

Progress Check 1

ANSWERS

Units in this pack

- Number 1
- Algebra 1

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 1 · Working with fractions (Lesson 1.1)

Q	Working	Answer
adding fractions		
1.1	$\frac{12}{20} + \frac{5}{20} = \frac{17}{20}$	$\frac{17}{20}$
1.2	$\frac{20}{24} + \frac{3}{24} = \frac{23}{24}$	$\frac{23}{24}$
1.3	$\frac{21}{30} + \frac{5}{30} = \frac{26}{30}$ $= \frac{13}{15}$	$\frac{13}{15}$
1.4	$\frac{6}{21} + \frac{7}{21} = \frac{13}{21}$	$\frac{13}{21}$
1.5	$\frac{9}{18} + \frac{4}{18} = \frac{13}{18}$	$\frac{13}{18}$
1.6	$\frac{9}{24} + \frac{8}{24} = \frac{17}{24}$	$\frac{17}{24}$
multiplying fractions		
1.7	$= \frac{15}{60}$ $= \frac{1}{4}$	$\frac{1}{4}$
1.8	$= \frac{28}{84}$ $= \frac{1}{3}$	$\frac{1}{3}$
1.9	$= \frac{30}{80}$ $= \frac{3}{8}$	$\frac{3}{8}$
1.10	$= \frac{6}{36}$ $= \frac{1}{6}$	$\frac{1}{6}$

1.11	$= \frac{20}{120}$ $= \frac{1}{6}$	$\frac{1}{6}$
1.12	$= \frac{21}{126}$ $= \frac{1}{6}$	$\frac{1}{6}$
dividing fractions		
1.13	$\frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$ $= 1\frac{7}{8}$	$1\frac{7}{8}$
1.14	$\frac{7}{8} \times \frac{3}{1} = \frac{21}{8}$ $= 2\frac{5}{8}$	$2\frac{5}{8}$
1.15	$\frac{9}{10} \times \frac{5}{3} = \frac{45}{30}$ $= \frac{3}{2}$ $= 1\frac{1}{2}$	$1\frac{1}{2}$
1.16	$\frac{5}{6} \times \frac{9}{4} = \frac{45}{24}$ $= \frac{15}{8}$ $= 1\frac{7}{8}$	$1\frac{7}{8}$
1.17	$\frac{2}{3} \times \frac{7}{4} = \frac{14}{12}$ $= \frac{7}{6}$ $= 1\frac{1}{6}$	$1\frac{1}{6}$
1.18	$\frac{8}{9} \times \frac{3}{2} = \frac{24}{18}$ $= \frac{4}{3}$ $= 1\frac{1}{3}$	$1\frac{1}{3}$
fractions of amounts		

1.19	$42 \div 6 = 7$ $7 \times 5 = 35$	**35**
1.20	$81 \div 9 = 9$ $9 \times 2 = 18$	**18**
1.21	$75 \div 5 = 15$ $15 \times 4 = 60$	**60**
1.22	$56 \div 7 = 8$ $8 \times 3 = 24$	**24**
1.23	$96 \div 8 = 12$ $12 \times 5 = 60$	**60**
1.24	$57 \div 3 = 19$ $19 \times 2 = 38$	**38**
fractions, decimals and percentages		
1.25	**(a)** $2 \div 5 = 0.4$ **(b)** $0.4 \times 100 = 40$	**(a) 0.4** **(b) 40%**
1.26	**(a)** $7 \div 20 = 0.35$ **(b)** $0.35 \times 100 = 35$	**(a) 0.35** **(b) 35%**
1.27	**(a)** $5 \div 8 = 0.625$ **(b)** $0.625 \times 100 = 62.5$	**(a) 0.625** **(b) 62.5%**
1.28	**(a)** $3 \div 4 = 0.75$ **(b)** $0.75 \times 100 = 75$	**(a) 0.75** **(b) 75%**
1.29	**(a)** $9 \div 25 = 0.36$ **(b)** $0.36 \times 100 = 36$	**(a) 0.36** **(b) 36%**
1.30	**(a)** $1 \div 8 = 0.125$ **(b)** $0.125 \times 100 = 12.5$	**(a) 0.125** **(b) 12.5%**

Number 1 · Order of operations and rounding (Lessons 1.2, 1.3)

Q	Working	Answer
order of operations		
2.1	**(a)** $4 \times 5 = 20$ $7 + 20 = 27$ **(b)** $6^2 = 36$ $36 \div 4 = 9$	**(a) 27** **(b) 9**
2.2	**(a)** $3 \times 6 = 18$ $20 - 18 = 2$ **(b)** $7^2 = 49$ $49 - 9 = 40$	**(a) 2** **(b) 40**
2.3	**(a)** $18 \div 3 = 6$ $12 + 6 = 18$ **(b)** $8^2 = 64$ $64 \div 8 = 8$	**(a) 18** **(b) 8**
2.4	**(a)** $5 \times 3 = 15$ $6 + 15 = 21$ **(b)** $6^2 = 36$ $36 \div 9 = 4$	**(a) 21** **(b) 4**
2.5	**(a)** $4 \times 5 = 20$ $30 - 20 = 10$ **(b)** $7^2 = 49$ $49 - 10 = 39$	**(a) 10** **(b) 39**
2.6	**(a)** $24 \div 4 = 6$ $8 + 6 = 14$ **(b)** $8^2 = 64$ $64 \div 4 = 16$	**(a) 14** **(b) 16**
rounding		
2.7		**(a) 5.3** **(b) 5.27**

2.8		**(a) 18.5** **(b) 18.5**
2.9		**(a) 9.1** **(b) 9.08**
2.10		**(a) 3.7** **(b) 3.66**
2.11		**(a) 27.4** **(b) 27.4**
2.12		**(a) 5.0** **(b) 5.00**

Algebra 1 · Substitution and simplifying (Lessons 2.1, 2.2, 2.3)

Q	Working	Answer
substitution		
3.1	**(a)** $3 \times 4 + (-1) = 11$ **(b)** $4^2 - 2 \times (-1) = 18$	**(a) 11** **(b) 18**
3.2	**(a)** $5 \times 2 + (-5) = 5$ **(b)** $2^2 - 1 \times (-5) = 9$	**(a) 5** **(b) 9**
3.3	**(a)** $2 \times 6 + (-2) = 10$ **(b)** $6^2 - 3 \times (-2) = 42$	**(a) 10** **(b) 42**
3.4	**(a)** $2 \times 5 + (-3) = 7$ **(b)** $5^2 - 4 \times (-3) = 37$	**(a) 7** **(b) 37**
3.5	**(a)** $5 \times 3 + (-4) = 11$ **(b)** $3^2 - 2 \times (-4) = 17$	**(a) 11** **(b) 17**
3.6	**(a)** $2 \times 7 + (-1) = 13$ **(b)** $7^2 - 5 \times (-1) = 54$	**(a) 13** **(b) 54**
simplifying		
3.7		**(a)** $3x + 7y$ **(b)** $18ab$

3.8		**(a)** $4p + 4q$ **(b)** $14cd$
3.9		**(a)** $3a + 5b$ **(b)** $20mn$
3.10		**(a)** $5x + 5y$ **(b)** $24pq$
3.11		**(a)** $2m + 3n$ **(b)** $14xy$
3.12		**(a)** $4c + 5d$ **(b)** $24ab$
expanding brackets		
3.13	**(a)** $4 \times 3x = 12x$ $4 \times (-5) = -20$ **(b)** $x \times x = x^2$ $x \times 6 = 6x$	**(a)** $12x - 20$ **(b)** $x^2 + 6x$
3.14	**(a)** $6 \times 2x = 12x$ $6 \times 1 = 6$ **(b)** $x \times x = x^2$ $x \times (-3) = -3x$	**(a)** $12x + 6$ **(b)** $x^2 - 3x$
3.15	**(a)** $2 \times 5x = 10x$ $2 \times (-4) = -8$ **(b)** $x \times x = x^2$ $x \times 9 = 9x$	**(a)** $10x - 8$ **(b)** $x^2 + 9x$
3.16	**(a)** $7 \times 2x = 14x$ $7 \times (-3) = -21$ **(b)** $x \times x = x^2$ $x \times 8 = 8x$	**(a)** $14x - 21$ **(b)** $x^2 + 8x$

3.17	**(a)** $5 \times 3x = 15x$ $5 \times 1 = 5$ **(b)** $x \times x = x^2$ $x \times (-2) = -2x$	**(a)** $15x + 5$ **(b)** $x^2 - 2x$
3.18	**(a)** $3 \times 4x = 12x$ $3 \times (-7) = -21$ **(b)** $x \times x = x^2$ $x \times 1 = 1x$	**(a)** $12x - 21$ **(b)** $x^2 + x$
multiplying the same letter		
3.19	$3 \times 5 = 15$ $m \times m = m^2$	$15m^2$
3.20	$4 \times 2 = 8$ $x \times x = x^2$	$8x^2$
3.21	$6 \times 1 = 6$ $y \times y = y^2$	$6y^2$
3.22	$5 \times 4 = 20$ $a \times a = a^2$	$20a^2$
3.23	$2 \times 9 = 18$ $k \times k = k^2$	$18k^2$
3.24	$7 \times 3 = 21$ $n \times n = n^2$	$21n^2$

Algebra 1 · Solving equations (Lesson 2.4)

Q	Working	Answer
two-step equations		
4.1	$5x = 35$ $x = 7$	$x = 7$
4.2	$2x = 16$ $x = 8$	$x = 8$

4.3	$6x = 36$ $x = 6$	$x = 6$
4.4	$4x = 28$ $x = 7$	$x = 7$
4.5	$7x = 42$ $x = 6$	$x = 6$
4.6	$3x = 21$ $x = 7$	$x = 7$
equations with brackets		
4.7	$5x - 15 = 20$ $5x = 35$ $x = 7$	$x = 7$
4.8	$2x + 12 = 22$ $2x = 10$ $x = 5$	$x = 5$
4.9	$4x - 4 = 28$ $4x = 32$ $x = 8$	$x = 8$
4.10	$3x + 15 = 27$ $3x = 12$ $x = 4$	$x = 4$
4.11	$6x - 12 = 30$ $6x = 42$ $x = 7$	$x = 7$
4.12	$2x - 14 = 10$ $2x = 24$ $x = 12$	$x = 12$
unknowns on both sides		

4.13	$3x - 2 = 13$ $3x = 15$ $x = 5$	$x = 5$
4.14	$5x + 1 = 26$ $5x = 25$ $x = 5$	$x = 5$
4.15	$5x - 7 = 8$ $5x = 15$ $x = 3$	$x = 3$
4.16	$4x + 4 = 24$ $4x = 20$ $x = 5$	$x = 5$
4.17	$4x - 5 = 11$ $4x = 16$ $x = 4$	$x = 4$
4.18	$3x + 3 = 18$ $3x = 15$ $x = 5$	$x = 5$
equations with x over a number		
4.19	$\frac{x}{4} = 5$ $x = 5 \times 4$ $x = 20$	$x = 20$
4.20	$\frac{x}{5} = 6$ $x = 6 \times 5$ $x = 30$	$x = 30$
4.21	$\frac{x}{2} = 7$ $x = 7 \times 2$ $x = 14$	$x = 14$

4.22	$\frac{x}{6} = 3$ $x = 3 \times 6$ $x = 18$	$x = 18$
4.23	$\frac{x}{3} = 5$ $x = 5 \times 3$ $x = 15$	$x = 15$
4.24	$\frac{x}{5} = 6$ $x = 6 \times 5$ $x = 30$	$x = 30$